

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results by over 2 BLEU. On the WMT 2014 English-to-French task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs.

1. Introduction

Recurrent neural networks, particularly long short-term memory and gated recurrent neural networks, have been firmly established as state-of-the-art approaches in sequence modeling. Recurrent models typically factor computation along the symbol positions of the input and output sequences, precluding parallelization within training examples.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

3. Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure. The encoder maps an input sequence of symbol representations to a sequence of continuous representations. Given this, the decoder then generates an output sequence one element at a time, consuming the previously generated symbols as additional input when generating the next.

3.1 Encoder and Decoder Stacks. The encoder is composed of a stack of $N=6$ identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism, and a position-wise fully connected feed-forward network. We employ a residual connection around each of the two sub-layers, followed by layer normalization.

3.2 Attention. An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

$$\text{Eq: } \text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

3.2.2 Multi-Head Attention. Instead of performing a single attention function with d_{model} -dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections. In this work we employ $h=8$ parallel attention layers, or heads. For each of these we use $d_k = d_v = d_{\text{model}}/h = 64$.

3.5 Positional Encoding. Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens. We use sine and cosine functions of different frequencies for this purpose.

5. Training

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. We used the Adam optimizer with $\beta_1=0.9$, $\beta_2=0.98$ and $\epsilon=1e-9$, varying the learning rate over the course of training according to a warmup-then-decay schedule with $\text{warmup_steps}=4000$.

6. Results

On the WMT 2014 English-to-German translation task, the big Transformer model outperforms the best previously reported models, including ensembles, by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. Even our base model surpasses all previously published models and ensembles, at a fraction of the training cost of any of the competitive models.

7. Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention. For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers.